

Your Full Name

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May 28, 2026

Prof. Recipient Name
Search Committee Chair
Department of Applied Mathematics
Target University or Company
City, State, ZIP

Re: Application for Postdoctoral Researcher in Numerical Analysis (Position #12345)

Dear Prof. Recipient Name,

I am writing to apply for the postdoctoral position in numerical analysis at Target University. I am completing my Ph.D. in Applied Mathematics at the University of Somewhere under the supervision of Prof. Jane Doe, with an expected graduation of May 2026. My research sits at the intersection of *numerical methods for stochastic PDEs*, *uncertainty quantification*, and *high-performance scientific computing* — areas directly aligned with your group's work on multilevel Monte Carlo methods for subsurface flow problems.

Dissertation research. My dissertation develops *adaptive* multilevel Monte Carlo (MLMC) estimators for elliptic stochastic PDEs $-\nabla \cdot (a(\mathbf{x}, \omega) \nabla u) = f$, where a is a log-normal random field. Standard MLMC requires pre-specified mesh hierarchies and sample counts, which is fragile when the coefficient is spatially rough. I designed an online algorithm that selects level parameters by minimising a cost-versus-variance trade-off estimated from a Gaussian process surrogate, and proved optimal $\mathcal{O}(\varepsilon^{-2} |\log \varepsilon|^2)$ complexity without *a priori* knowledge of coefficient regularity. Implemented in C++/MPI with FEniCS and PETSc, the method achieves a $3\times$ wall-clock speedup over standard MLMC on a 3-D Darcy flow benchmark with weak scaling to 512 cores. This work will appear in the *SIAM Journal on Scientific Computing* (2024).

Optimization and machine learning. During a research internship at National Lab, I proved sharp $\mathcal{O}(k^{-1/2})$ convergence rates for stochastic gradient descent under (L, σ) -smoothness — strictly weaker than the standard bounded-gradient assumption — and showed the rate is unimprovable by constructing matching lower bounds. This preprint is under revision for *NeurIPS 2024* and informs my long-term interest in theoretically grounded optimisation for large-scale inverse problems.

Why Target University. I am particularly drawn to Prof. Recipient's recent work on [specific paper or project]. The questions you raise about [specific technical aspect] connect directly to the algorithmic ideas I have developed, and I see clear potential for collaboration on [concrete joint direction]. I also look forward to contributing to graduate teaching; I have served as instructor of record for Calculus II and am keen to develop courses in numerical methods or scientific computing.

I have enclosed my curriculum vitae, project portfolio, and three letters of reference from Prof. Jane Doe, Dr. John Smith, and Prof. Alice Brown. I would welcome the opportunity to discuss my research at your convenience.

Sincerely,

Your Full Name

Ph.D. Candidate, Applied Mathematics
University of Somewhere
you@university.edu